

GARV ARORA

Robotics & Embedded Systems Engineer | Autonomous Systems | ADAS

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SUMMARY

R&D Lead overseeing 2 autonomous systems teams (50+ engineers) at SEDS VIT — apex body of SEDS India. Final-year CS undergraduate (IoT) at VIT with 2+ years developing ROS2 navigation stacks, CUDA-accelerated perception pipelines, mmWave sensor fusion, and ARM bare-metal RTOS. Placed 12th globally at IRC 2025 (100+ teams). Targeting roles in robotics software, embedded systems, ADAS, or autonomous vehicle engineering.

EDUCATION

Vellore Institute of Technology | Vellore, India

Bachelor of Technology – Computer Science & Engineering (Specialization: IoT)

CGPA: 8.37 / 10.0

August 2023 – July 2027

Gems International School | Gurugram, India

Central Board of Secondary Education (CBSE)

Class 10: 89.3% | Class 12: 76.4%

November 2016 – April 2023

TECHNICAL SKILLS

Languages: C, C++, Python, Java, JavaScript, SQL, Bash, Verilog, C#, HTML/CSS

Robotics & Autonomous Systems: ROS2, Nav2, MoveIt2, RTAB-Map, Micro-ROS, SLAM, EKF/UKF, PID Control, Point Cloud Processing, mmWave Radar

Perception & ML: OpenCV, YOLOv8, MediaPipe, Intel RealSense SDK, PyTorch, TensorFlow, scikit-learn, CUDA, HuggingFace Transformers

Embedded & Infrastructure: ARM Cortex-M, ESP32, I2C/SPI/UART, FreeRTOS, Docker, Git, GitLab CI/CD, Linux, AWS, PyQt, MongoDB

WORK EXPERIENCE

SEDS Projects VIT | SEDS India

January 2026 – Present

Research & Development Lead (Team Ardra – UAV | Team Vyadh – Rover)

Vellore, India

- Direct R&D roadmap for 2 autonomous systems teams (50+ engineers) — Team Ardra (UAV/drone) and Team Vyadh (planetary rover) — under SEDS India, the national apex body for space engineering students.
- Mentor sub-team leads on autonomous navigation stack architecture, embedded sensor integration, and UAV flight control, driving cross-team knowledge transfer ahead of IRC, IRDC, and UAV competitions.

Team Vyadh – Martian Rover | SEDS VIT

April 2024 – Present

Autonomous Systems Developer

Vellore, India

- Architected a full ROS2 navigation stack (Nav2, RTAB-Map Visual SLAM, RealSense D455) generating real-time 3D elevation maps, cutting manual teleoperation time by 70% in competition.
- Slashed CV pipeline latency by ~60% and raised throughput 4x via CUDA, YOLOv8, ArUco/AprilTag detection, and Linux V4L2 tuning across 4 simultaneous camera streams.
- Deployed Micro-ROS firmware to 3 ESP32 nodes; fused IMU + encoder data with an EKF, reducing odometry drift by ~40% and achieving sub-10 cm autonomous positioning accuracy.
- Launched a PyQt5 mission dashboard with live telemetry, parameter tuning, and FSM sequencing — used by 5 operators across 3 competition missions with zero system crashes.

Wissen Baum Engineering Solutions

May 2025 – July 2025

Software Automation Intern

Pune, India

- Authored a BDD test suite (Python, Behave, pytest) covering 40+ scenarios, raising regression coverage from 0% to 100% on 3 core product flows in 6 weeks.
- Accelerated GitLab CI/CD by parallelizing 3 stages (Playwright + Cypress), trimming run time ~35% (18 min to 12 min); replaced manual QA with a pandas/NumPy script processing 10,000+ rows per run.

PROJECTS

mmWave Earthquake Survivor Detection Robot | ROS2, Python, TI IWR6843, ESP32

2026

- Interfaced a TI IWR6843 mmWave radar with ROS2 to detect survivor breathing signatures through 30 cm of rubble; fused radar point-cloud with IMU odometry for autonomous debris navigation, flagging coordinates with <15 cm error.

Gesture-Controlled Robotic Arm | C++, Arduino, MPU6050, Flex Sensors

2026

- Mapped real-time MPU6050 orientation and 5-finger flex sensor input to 6-DOF servo commands with <50 ms latency; trained gesture classifier on 200+ samples, achieving 94% accuracy across 8 gestures.

HayaiOS – Bare-Metal RTOS | C, ARM Assembly, ARM Cortex-M4

2026

- Implemented preemptive scheduler (1 ms tick), context switching in <50 assembly instructions, mutex/semaphore primitives, and a full HAL isolating all register-level I/O on ARM Cortex-M4.

3D Reconstruction – IMU-Enhanced KinectFusion | C++, OpenCV, Intel RealSense SDK

2025

- Stabilized KinectFusion TSDF reconstruction by fusing RealSense D455 depth and IMU via complementary filter, reducing tracking failures by ~50% across a 6-DOF fast-motion test trajectory.

CERTIFICATIONS & ACHIEVEMENTS

IRC 2025: Ranked 12th globally (100+ international teams) | **IRDC 2025:** Top 5 Finalist | **Oracle Cloud Infrastructure 2025:** Generative AI Professional | Data Science Professional